

1. Executive Summary & Problem Space

The system features a hybrid dual-interface architecture: (1) a lightweight, responsive **Server-Sent Events (SSE) streaming text chat** grounded by an embedded in-memory CSV-based RAG engine with TF-IDF indexing; and (2) an ultra-low-latency **real-time bidirectional voice assistant** built directly on Google's gemini-2.5-flash-native-audio-latest model via WebSockets with seamless client-side tool calling and barge-in audio interruption.

2. Technical Stack Specification

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3. Frontend Architecture & Voice Pipeline

A. Modular UI Component Hierarchy

The client architecture adheres to a clean separation of concerns, orchestrated through VoiceAssistantApp.tsx:

- **Chat Assistant (ChatAssistant.tsx):** Real-time streaming conversation container receiving SSE events via chatService.ts. Includes multi-session history, quick-prompt suggestions, inline scheme previews, and Markdown rendering.
- **Voice Talking Screen (VoicePanel.tsx):** High-contrast interface featuring live state transitions ('Your turn — speak now', 'Gram Sathi is speaking...'), localized across 14 languages, turn indicators, and barge-in controls.
- **Crop Calendar (CropCalendar.tsx):** Interactive agricultural timeline with StateSelector, CropSelector, and MonthCalendarGrid. Features multi-season scheduling (Kharif, Rabi, Zaid) and sowing/growing/harvesting windows.
- **Service Detail Page (ServiceDetailPage.tsx):** Dedicated, bookmarkable, and shareable routes (/services/:slug) detailing marquee schemes with a bilingual English/Hindi toggle.

B. Real-Time Gemini Live Audio Engine

Rather than relying on basic browser speech recognition, the voice mode utilizes Google's Gemini Live native audio streaming pipeline:

- **Microphone Capture (useMicrophone.ts):** Ingests user voice at 16kHz via Web Audio API, calculates RMS energy for local voice activity detection (VAD), and converts samples to linear 16-bit PCM.
- **Audio Transcoding (audioProcessor.ts):** Zero-copy conversion between Float32Array and Int16Array, linear interpolation downsampler, and Base64 wire encoder.
- **Audio Playback & Barge-In (useAudioPlayback.ts):** Streams 24kHz PCM chunks through an AudioContext queue. If user speech is detected during AI speech, it immediately cancels playback and resets audio buffers (barge-in interruption).
- **Token Pre-Warming (tokenService.ts):** Caches and pre-fetches ephemeral tokens on page hover/load so the WebSocket session connects in under 300ms.
- **Fluid Canvas Visualization (AnimatedGlobe.tsx):** High-DPI HTML5 Canvas rendering a fluid, 3D-like orb responding to live frequency data and mic RMS energy.

C. Client-Side Function Calling Matrix (toolManager.ts)

Voice Tool Function	Trigger Condition	Action Executed on Client/Server
getCropCalendar	Farmer asks about sowing or harvesting	Queries /api/crop-calendar and returns concise spoken summary.
setCropCalendarState	User mentions their state or location	Synchronously updates the calendar UI state and context.
navigateToScheme	User asks to open or apply for a scheme	Dispatches route navigation to /services/:slug.
searchScheme	User asks about loans, subsidy, or bima	Invokes scheme search API and returns top matches.
showDocuments	User asks what documents are required	Pops up document checklist modal for the active scheme.

4. Backend API & In-Memory RAG Engine

A. Secure Ephemeral Token Minting (liveToken.ts)

To permit browser clients to establish WebSocket connections directly with Google's Gemini Live infrastructure without revealing credentials, the backend generates short-lived authentication tokens:

1. Client requests a token via GET /api/live/token on startup or button hover.
2. Server invokes ai.authTokens.create({ config: { expireTime: +30min } }) using the secret GEMINI_API_KEY.
3. A restricted, single-use token identifier (e.g., auth_tokens/...) is returned. The master key never leaves the server.

B. In-Memory RAG Engine & Multilingual Synonyms (knowledgeBase.ts)

The backend implements an embedded, zero-latency RAG pipeline that avoids the cost and latency of external vector databases:

- **CSV Ingestion:** Loads data/knowledge.csv (870+ scheme and cooperative records) at startup using csv-parse/sync.
- **Synonym Expansion Matrix:** Maps vernacular terminology to standard government terms across Hindi, Hinglish, and English. For instance, 'karj', 'rin', and 'credit' expand to 'KCC' and 'loan'; 'bima' expands to 'PMFBY' and 'fasal insurance'.
- **TF-IDF Scorer:** Calculates token frequency / inverse document frequency across categories, titles, keywords, questions, and answers to retrieve the top 2 to 5 matching records.

C. Resilient AI Orchestration & Fallbacks (gemini.ts)

All model inferences pass through an automated fallback waterfall to guarantee 100% uptime: gemini-3.6-flash → gemini-3.7-flash → gemini-3.5-flash-lite → gemini-flash-latest. If a model encounters a transient 503 high-demand error, it executes exponential backoff retry before falling back to the next candidate.

Additionally, normalizeSchemeUrls scans responses and replaces deprecated portal links (such as legacy kviconline or pmkusum URLs) with verified, active myscheme.gov.in deep-links.

5. Security, Privacy & Data Governance

Security & Governance Domain	Implementation Details & Safeguards
API Key Security	The permanent GEMINI_API_KEY is stored strictly in server environment variables. Browser clients receive only short-lived ephemeral tokens with restricted session scopes.
Citizen Data Privacy	User session profiles and conversation histories are persisted strictly on client devices (sessionStorage / localStorage). The system explicitly forbids asking for or storing Aadhaar, OTP, Bank PIN, or passwords.
Zero Hallucination Policy	Strict system prompts instruct the model never to invent eligibility criteria, subsidy amounts, or application deadlines. Missing details trigger fallback to official verification advice.
14+ Language Detection	Regex-based Unicode script identification (Devanagari, Gurmukhi, Bengali, Telugu, Tamil, Kannada, Gujarati, Malayalam, Odia, Urdu) + Hinglish keyword matching guarantees immediate vernacular responses.

6. Architectural Summary & Conclusion

The Sahkar Sathi (Gram Sathi) codebase exemplifies modern, resilient AI architecture designed for the underserved rural demographic. By combining a zero-credential browser voice pipeline (Gemini Live Native Audio), an embedded in-memory RAG system with deep multilingual synonym awareness, and comprehensive crop data from the Government of India's UPaG portal, the platform delivers instant, trustworthy, and empathetic assistance to millions of Indian farmers and citizens.